

# MasterviewCEL Energy Solutions

## Solar System Sizing Engine & Pricing Model Audit

Document Ref: ENG-AUDIT-2026-Q1 | Date: 14/09/2026 | Scope: System Calibration

### 1. Executive Summary & Calibration Context

Following technical field review, the Solar Estimator calculation model was audited and calibrated to eliminate hyper-inflated estimates for residential and commercial off-grid installations in Nigeria.

Prior sizing calculated appliances assuming flat 24-hour continuous wattage without duty cycle adjustment, and defaulted to low system voltage (12V/24V). For a standard home profile (2 Fans, 1 Fridge, 20 Lights, 1HP Inverter AC on 24h backup), the previous system yielded an impractical 34.5 kWh/day, 2646 Ah battery bank, and N10.2M estimate. The calibrated model accurately reflects real-world thermal cycling, 48V architecture, and prevailing 2026 Nigerian turnkey commercial pricing.

### 2. Benchmark Test: 2 Fans, 1 Fridge, 20 Lights, 1 AC (1000W) - 24h Off-Grid

| Engineering Parameter | Previous Model (Uncalibrated) | Calibrated Model (2026 Standard) | Engineering Rationale                                                           |
|-----------------------|-------------------------------|----------------------------------|---------------------------------------------------------------------------------|
| Daily Energy Yield    | 34.5 kWh / day                | 12.1 kWh / day                   | Applies compressor thermostat duty cycle (55%) & daylight lighting cap.         |
| Inverter Capacity     | 2.0 kVA (Undersized)          | 3.5 kVA / 48V Pure Sine          | Mandatory 3.5kVA+ pure sine wave hybrid to handle inductive AC startup surge.   |
| Battery Storage       | 2646 Ah (12V/24V bus)         | 10.9 kWh (228 Ah @ 48V)          | Standardized on 48V bus; accounts for daytime direct PV power consumption.      |
| Solar PV Array        | 22 x 450W Panels              | 8 x 450W Panels (3.6 kW)         | Balanced series-parallel MPPT strings; ~16 m2 roof space requirement.           |
| Turnkey Investment    | NGN 10,212,048                | NGN 5,770,560                    | Prevailing Nigerian market EPC rates (Tier-1 Mono, LiFePO4, BoS, Installation). |

### 3. Appliance Duty Cycle & Energy Formula Matrix

| Appliance Category             | Rated Watts   | Effective Hours | Duty Cycle  | Effective Daily Wh | Operating Behavior                                                    |
|--------------------------------|---------------|-----------------|-------------|--------------------|-----------------------------------------------------------------------|
| Inverter Air Conditioner (1HP) | 1,000 W       | Max 12 h/day    | 0.55 (55%)  | 6,600 Wh           | Thermostat switches compressor on/off during afternoon heat & night.  |
| Refrigerator / Freezer         | 150 W         | 24 h/day        | 0.38 (38%)  | 1,368 Wh           | 24h connected, but cooling compressor only cycles ~35-40% of the day. |
| LED Lighting (20 Points)       | 200 W         | Max 8 h/day     | 1.00 (100%) | 1,600 Wh           | Lights are switched off during daylight; active 6pm-11pm & 5am-6am.   |
| Standing Fans (2 Units)        | 140 W         | 18 h/day        | 1.00 (100%) | 2,520 Wh           | Continuous afternoon and overnight ventilation.                       |
| Water Pump (Borehole)          | 750 W - 1.5kW | Max 2 h/day     | 1.00 (100%) | 1,500 Wh           | Intermittent pumping to fill overhead gravity tanks.                  |
| TV & Home Entertainment        | 120 W         | Max 12 h/day    | 0.90 (90%)  | 1,296 Wh           | Smart TV, soundbar, and satellite decoder usage.                      |
| Electric Iron / Microwave      | 1,200 W       | Max 1.5 h/day   | 0.80 (80%)  | 1,440 Wh           | High wattage intermittent heating element.                            |

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## Technical Sizing Formulas & Turnkey Cost Breakdown

Document Ref: ENG-AUDIT-2026-Q1 | Technical Equations & Commercial Pricing

### 4. Engineering Sizing Mathematical Formulations

| Component              | Engineering Formula                                                       | Design Criteria & Derating Factors                                                                                                   |
|------------------------|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Inverter Sizing        | $P_{inv} = \max(\text{Total\_Steady} * 1.25, \text{Max\_Surge} * 0.55)$   | 25% safety headroom. Heavy load / AC rule: minimum 3.5 kVA pure sine wave to handle compressor inductive surge.                      |
| Bus Voltage (V_bus)    | V_bus = 12V (<1.5kVA), 24V (1.5-2.5kVA), 48V (>=3kVA or AC)               | Enforces 48V DC bus on any AC system to prevent massive current (I = P/V) that causes cable heating and terminal melting.            |
| Solar Array Sizing     | $P_{array} = \text{Daily\_Wh} / (\text{PSH} * 0.82)$                      | Peak Sun Hours (PSH = 4.80h Nigeria average). 0.82 efficiency accounts for MPPT tracking, temperature coefficient & dust.            |
| Battery Storage Sizing | $C_{batt} \text{ (Wh)} = (\text{Daily\_Wh} * 0.65) / (\text{DoD} * 0.90)$ | Day direct solar fraction = 35%. Battery supplies night & cloudy backup (65%). DoD: 85% for Lithium LiFePO4, 50% for Tubular.        |
| DC Cable Conductor     | $\Delta V = (2 * L * I * \rho) / A \leq 2.5\%$                            | Double-insulated solar DC cable sized to guarantee voltage drop <= 2.5% over distance L (15m bungalow, 25m duplex, 40m multi-story). |

### 5. Calibrated Turnkey Cost Breakdown (Nigerian Market 2026)

| Bill of Quantities (BoQ) Item              | Unit Rate Basis (NGN)                                    | Test Profile Cost (3.5kVA / 10.9 kWh / 8 Panels) |
|--------------------------------------------|----------------------------------------------------------|--------------------------------------------------|
| Solar PV Modules (Tier-1 Monocrystalline)  | NGN 140,000 per 450W module                              | 8 x 450W Panels = NGN 1,120,000                  |
| Pure Sine Wave Hybrid MPPT Inverter        | 1.5kVA: 420k   2.5kVA: 620k   3.5kVA: 880k   5kVA: 1.25M | 3.5 kVA 48V Pure Sine Hybrid = NGN 880,000       |
| Lithium LiFePO4 Battery Storage Bank       | NGN 290,000 per kWh (or Tubular NGN 145,000/kWh)         | 10.9 kWh (228 Ah @ 48V) = NGN 3,173,760          |
| MPPT Solar Charge Controller               | Integrated into Hybrid Inverters >= 2.5 kVA              | NGN 0 (Built-in High-Voltage MPPT)               |
| Balance of System (BoS) & Safety Equipment | DC SPD + AC SPD + DC Breakers + PV Cable (NGN 2,800/m)   | 46m PV Cable + SPDs + Isolators = NGN 288,800    |
| Installation, Aluminum Rails & Earthing    | Rail Kits, Clamps, Grounding Rod + Labor: (Qty*16k)+180k | Mounting, Earth Rod & Labor = NGN 308,000        |
| TOTAL TURNKEY SYSTEM INVESTMENT            | Inclusive of all hardware, logistics, and commissioning  | NGN 5,770,560 (Payback: ~5.0 Years)              |

### 6. Implementation Architecture & Source Code Files

- Backend Calculation Service: backend/services/energyModel\_enhanced.js
- Frontend Offline Fallback Engine: src/hooks/useEstimation.ts (identical getDutyCycle & pricing logic)
- Client User Interface: src/components/AppResult.tsx (displays battery capacity primarily in kWh with secondary Ah)
- Conductor & Thermal Drop Helpers: src/utils/helpers.ts
- Production Status: Verified with 0 TypeScript/Lint errors, committed and pushed to branch main.